

Meeting Report

4/16

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Progress

上次Meeting未完成：

✗ 套件不相容，無法跑code。

本次Meeting進度：

- ✓ 成功處理套件不相容的問題。
- ✓ Code可以順利執行，成功利用Jetson Orin Nano的GPIO控制伺服馬達。
- ✓ 成功在real-time情境下實現人臉PID追蹤(x-axis)。
- ✓ 分析不同PID參數對於追蹤人臉的影響。

Aim

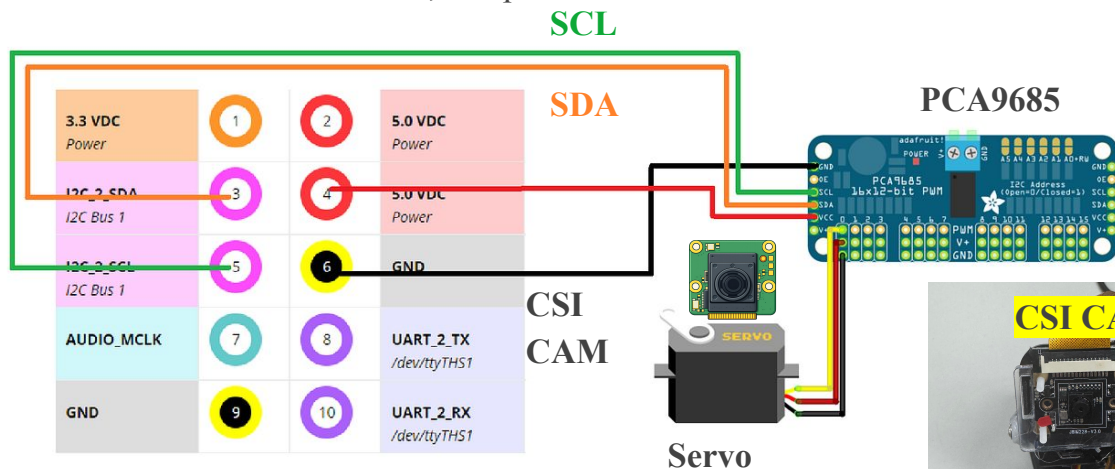
由於水下自由空間光通訊(UW-FSO)需精確對準，希望透過海中影像辨識船隻，使海底基站能準確對齊船上的 LED 通訊點，以確保穩定傳輸。



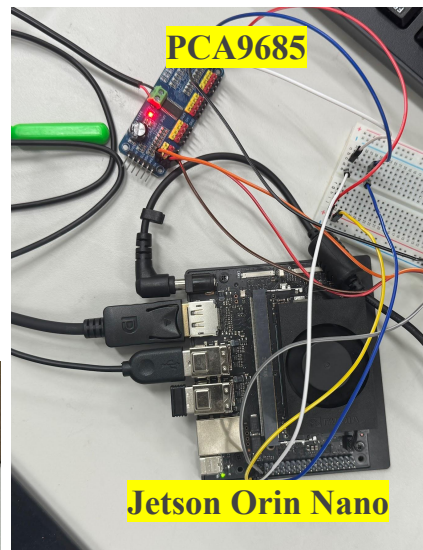
Experimental Setup

Jetson透過PCA9685模組，以I2C協議控制並輸出PWM訊號控制伺服馬達使CSI Camera跟著目標物轉動。

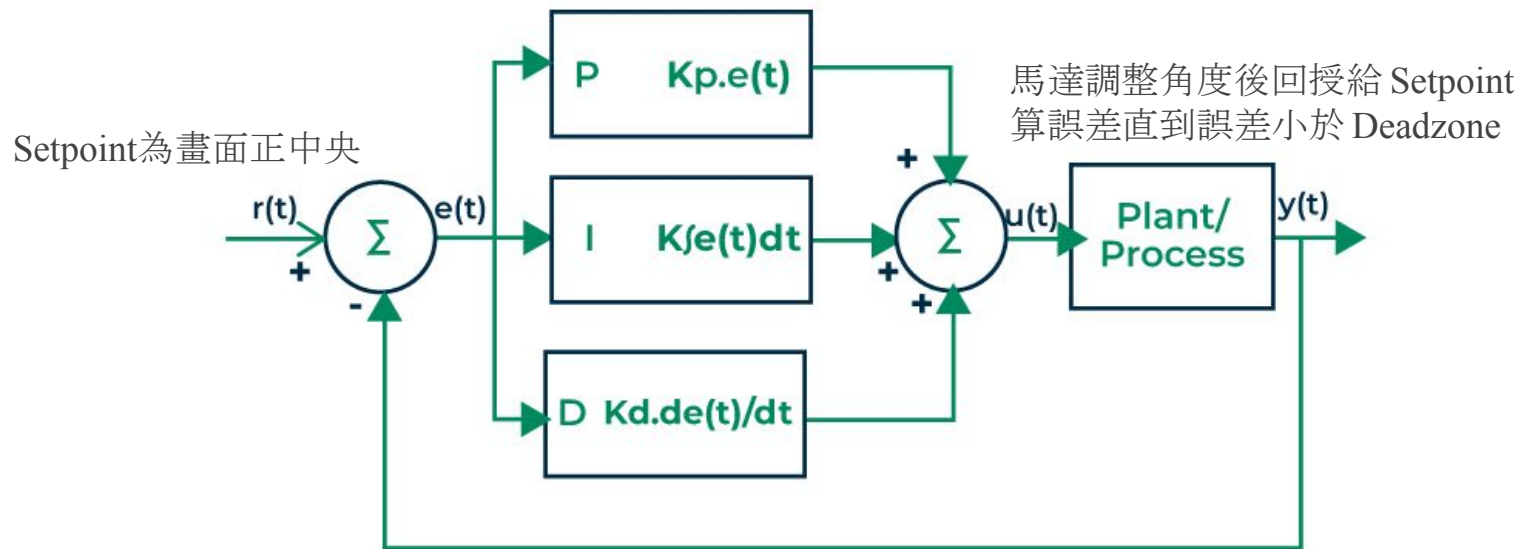
使用模型:YOLOv8,face.pt



Jetson Orin Nano

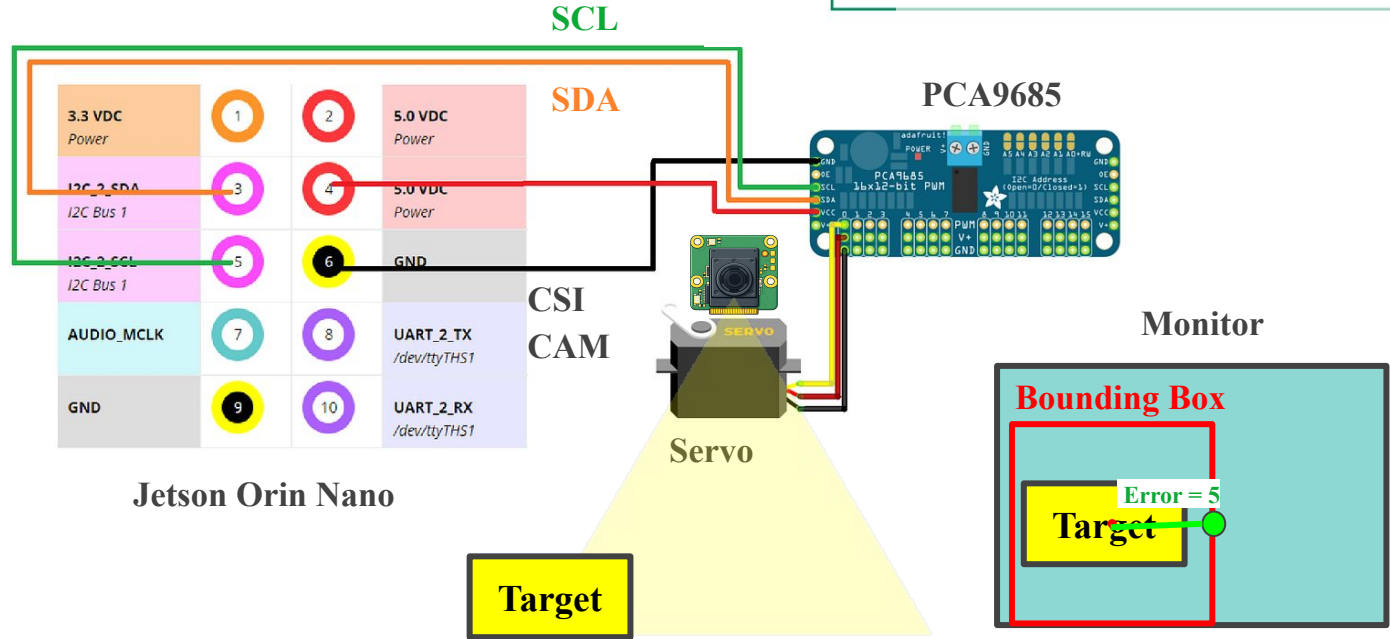
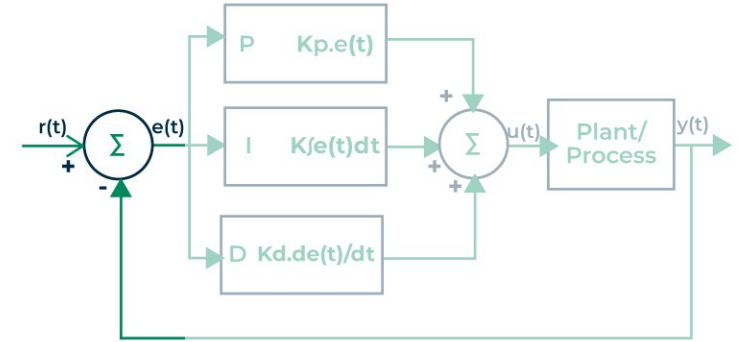


PID Control Flow



PID Control Flow

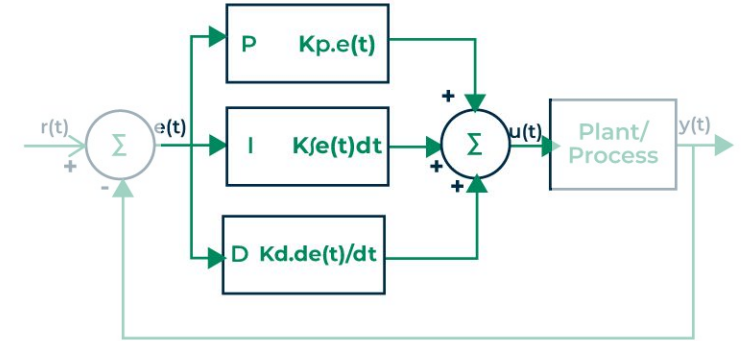
Error = $0 - (-5) = 5$



PID Control Flow

Error = $0 - (-5) = 5$

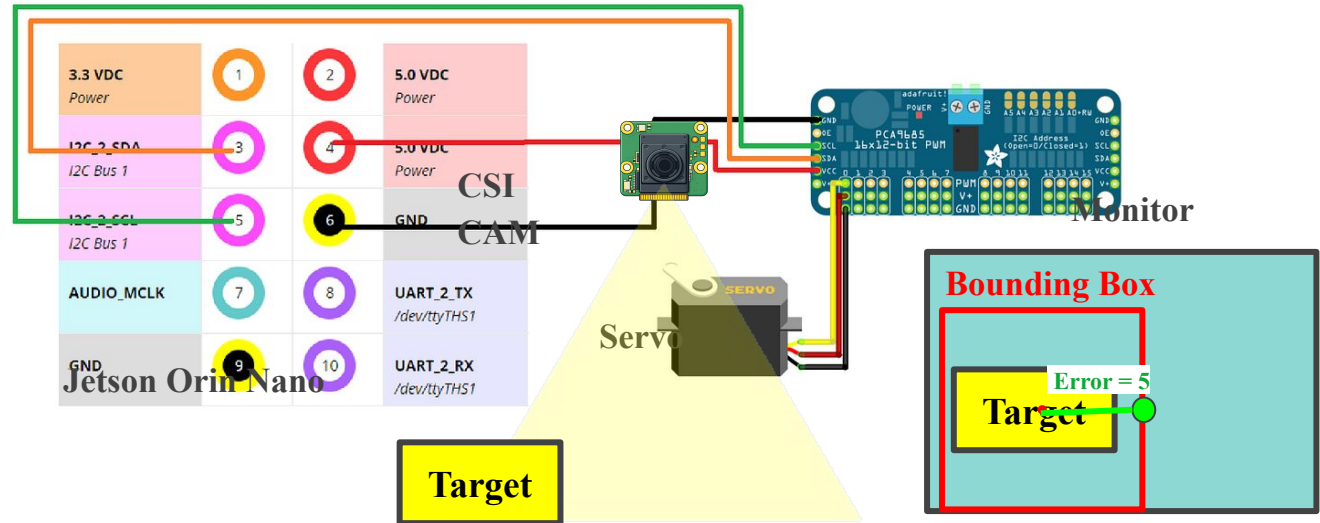
Servo_angle_new = $K_p \cdot \text{Error} + \text{Servo_angle}$



SCL

SDA

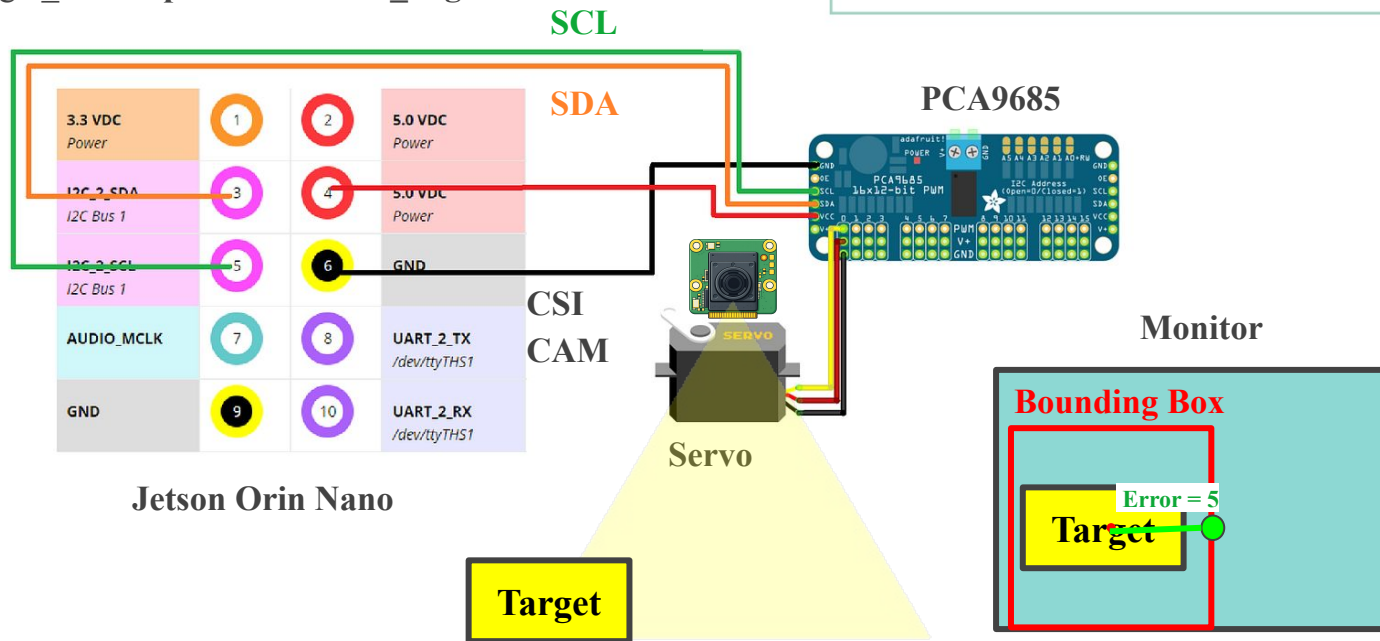
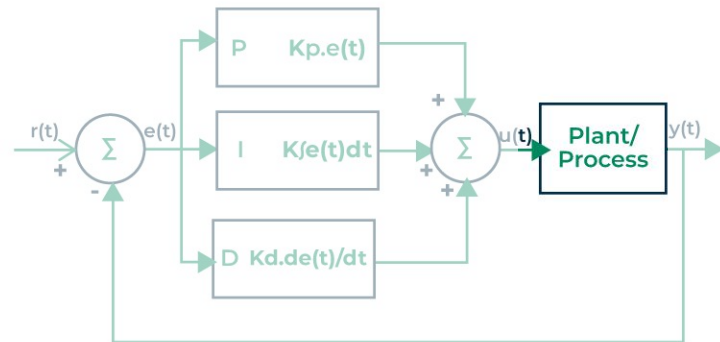
PCA9685



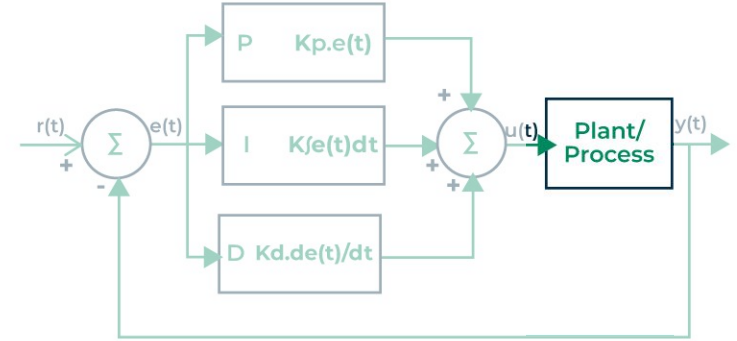
PID Control Flow

Error = $0 - (-5) = 5$

Servo_angle_new = $K_p \cdot \text{Error} + \text{Servo_angle}$

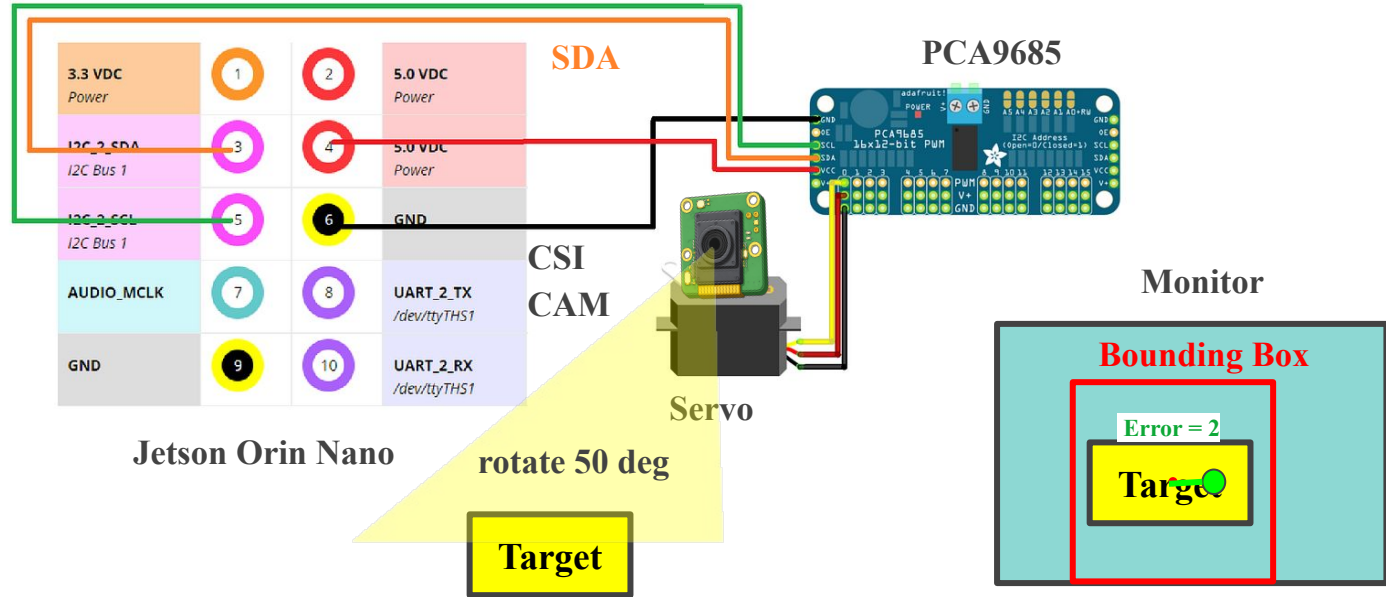


PID Control Flow



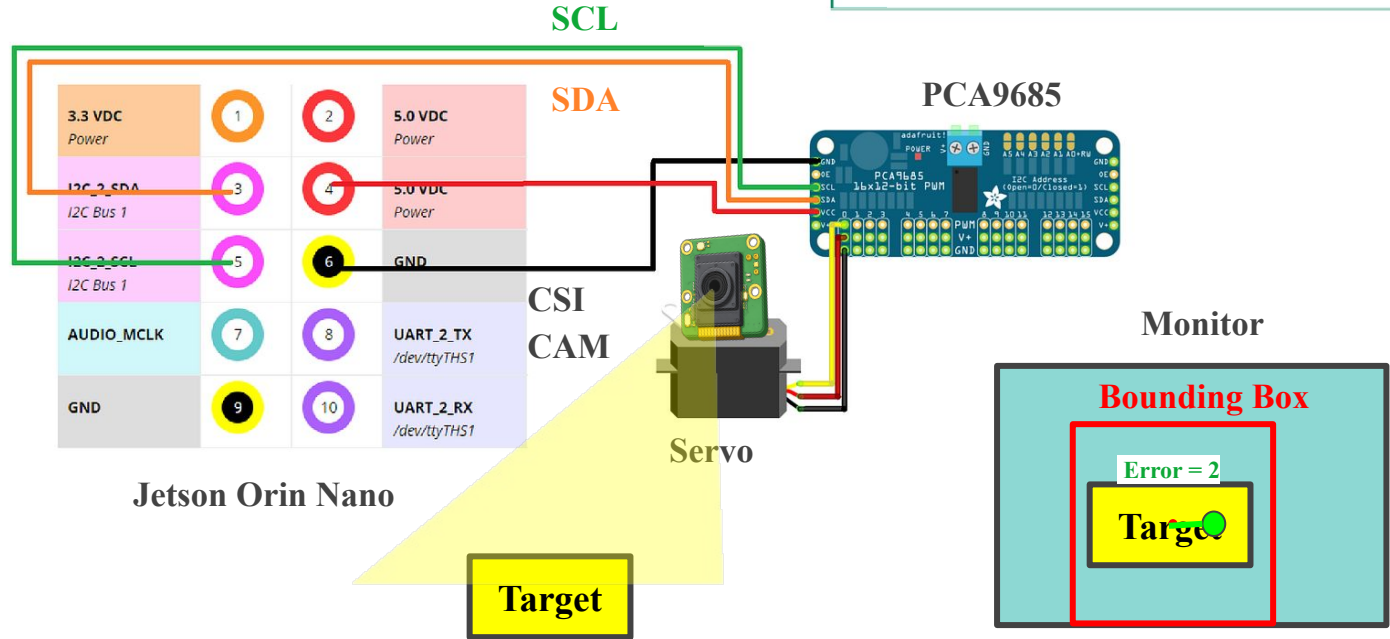
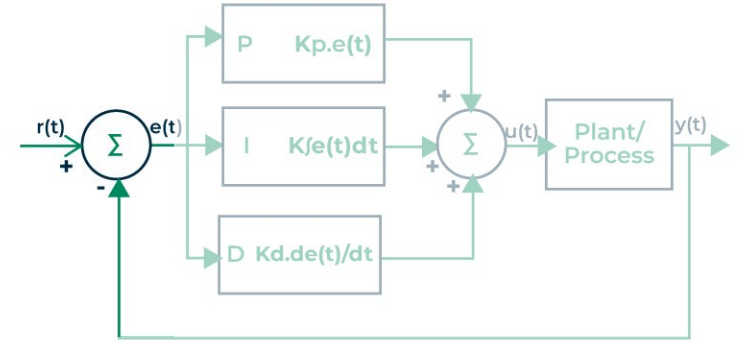
$$\text{Servo_angle_new} = K_p \cdot \text{Error} + \text{Servo_angle} = 10 \cdot 5 + 0 = 50$$

SCL



PID Control Flow

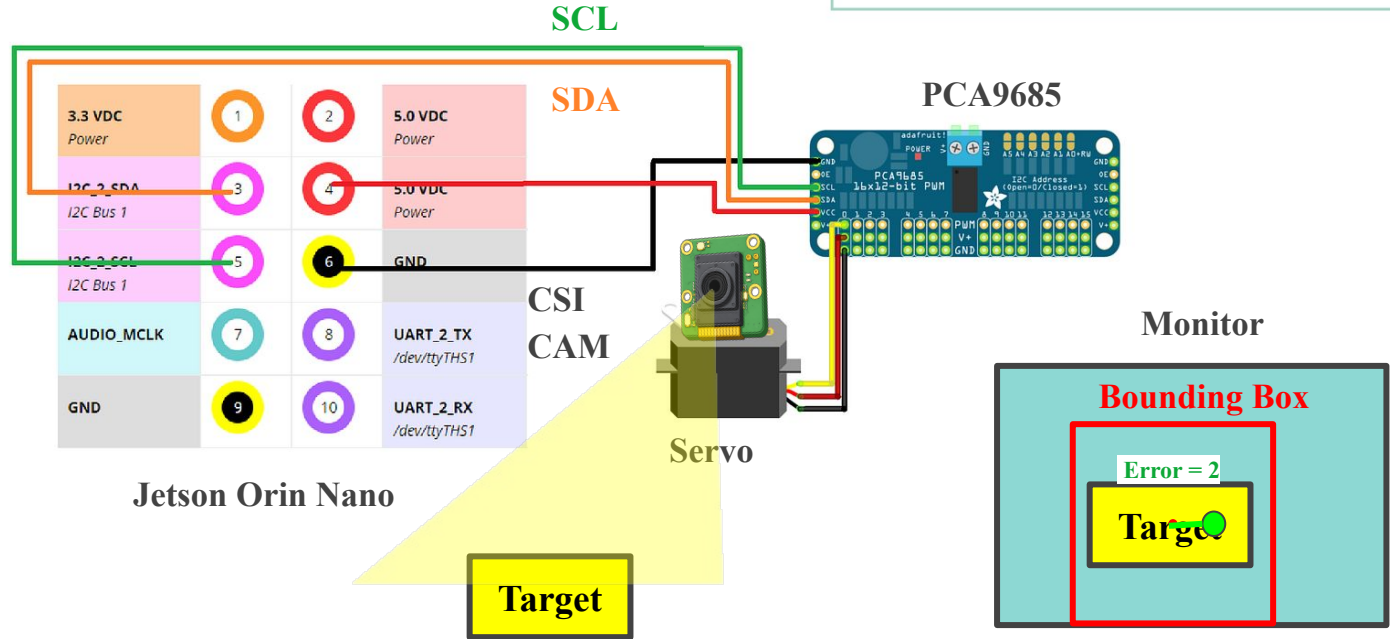
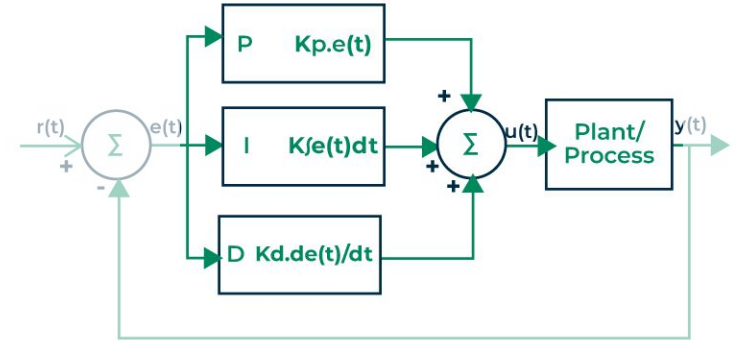
$$\text{Error} = 0 - (-2) = 2$$



PID Control Flow

$$\text{Error} = 0 - (-2) = 2$$

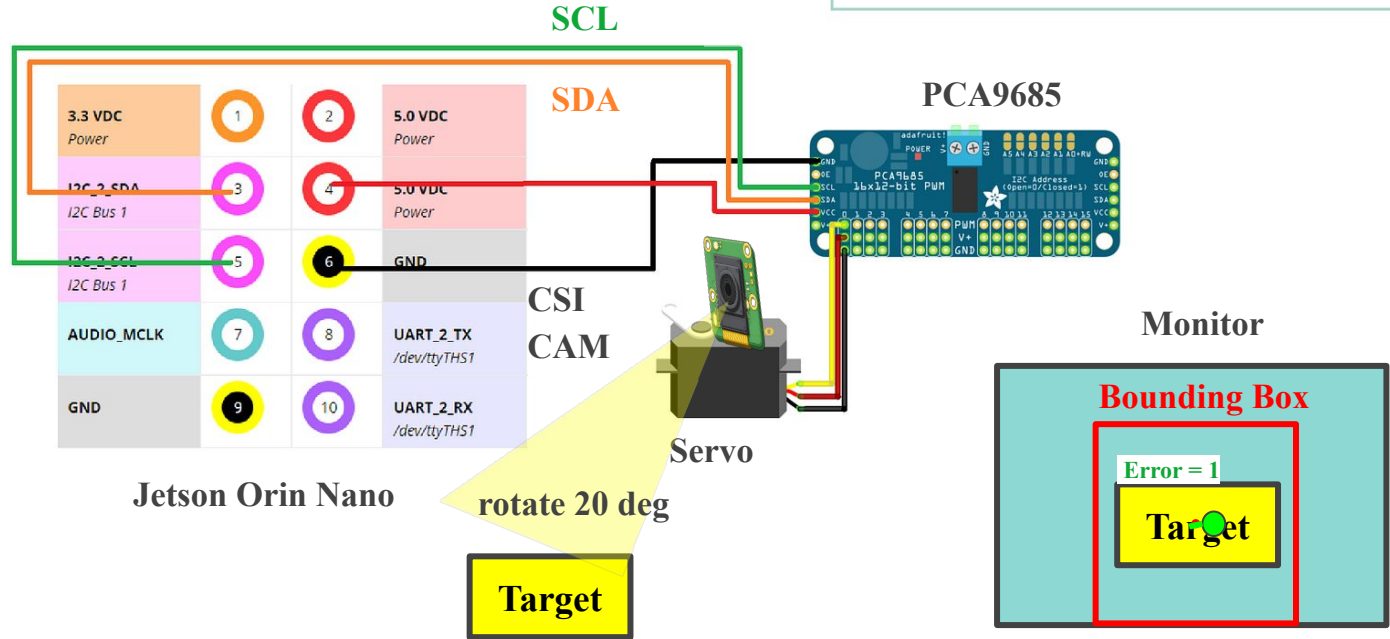
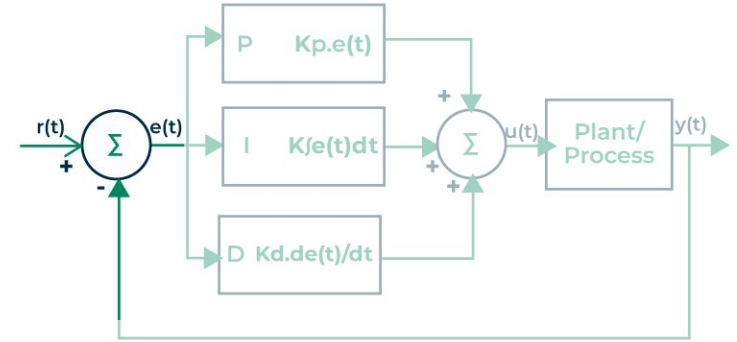
$$\text{Servo_angle_new} = K_p * \text{Error} + 50 = 10 * 2 + 50 = 70$$



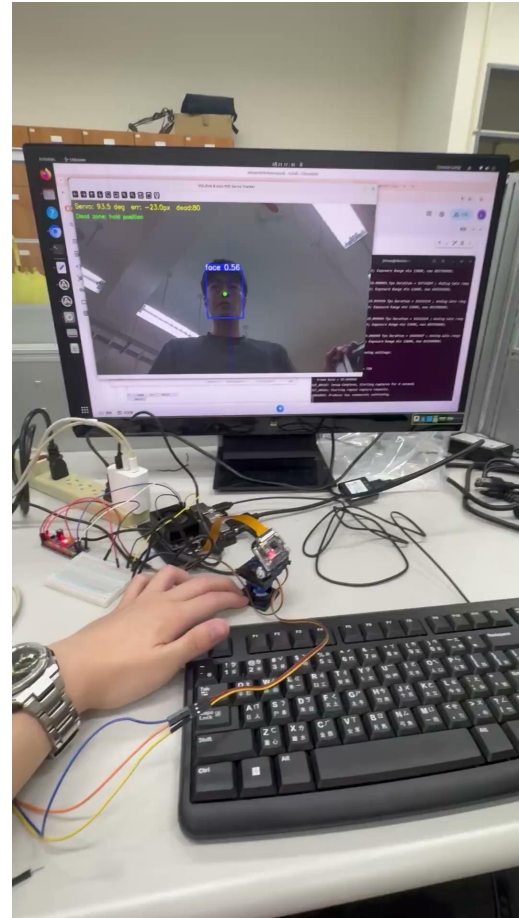
PID Control Flow

Error = $0 - (-1) = 1$

Stop when reaching dead zone



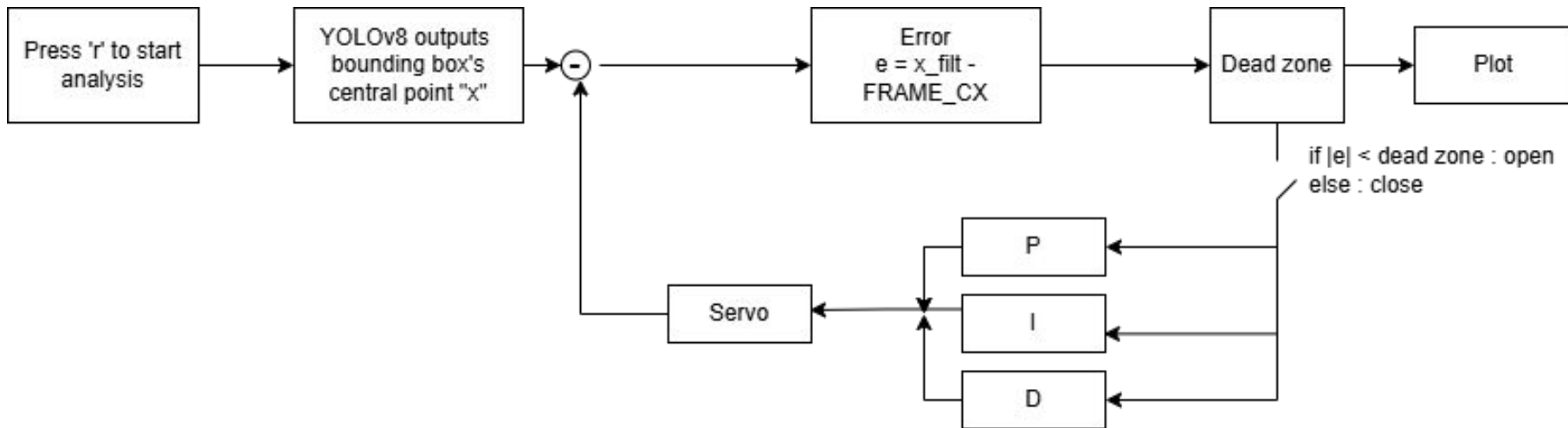
PID Control Flow



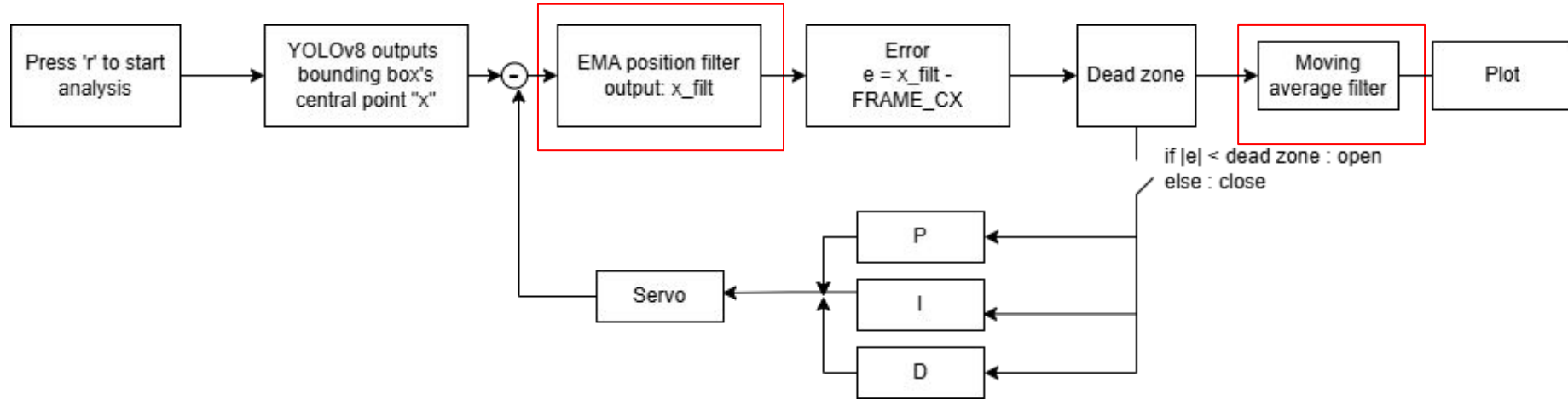


PID Control Flow

使用三組PID參數進行比較，按鍵盤上開始錄製。

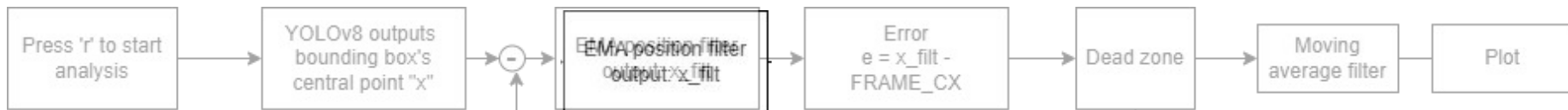


Applying Digital Filter





EMA Filter



$$y[n] = \alpha x[n] + (1 - \alpha)y[n - 1]$$

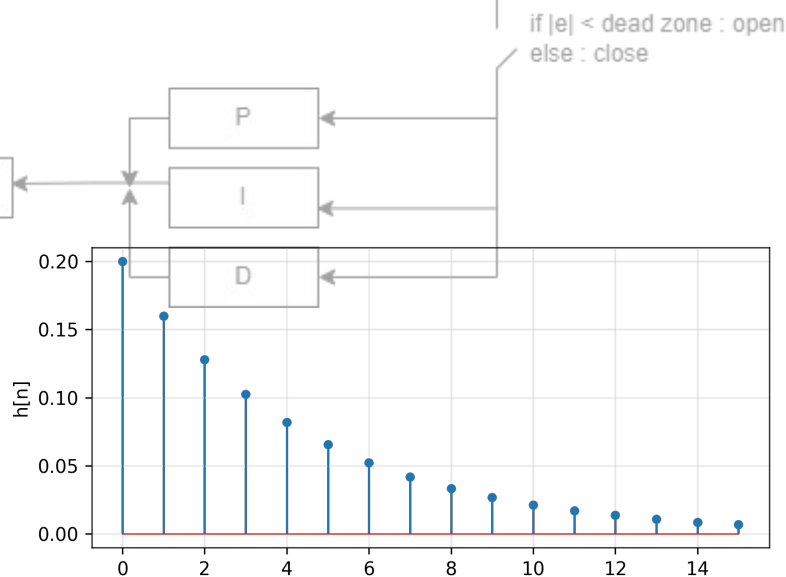
$$Y(z) = \alpha X(z) + (1 - \alpha)z^{-1}Y(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\alpha}{1 - (1 - \alpha)z^{-1}}$$

$$H(z) = \alpha + \alpha(1 - \alpha)z^{-1} + \alpha(1 - \alpha)^2z^{-2} + \dots$$

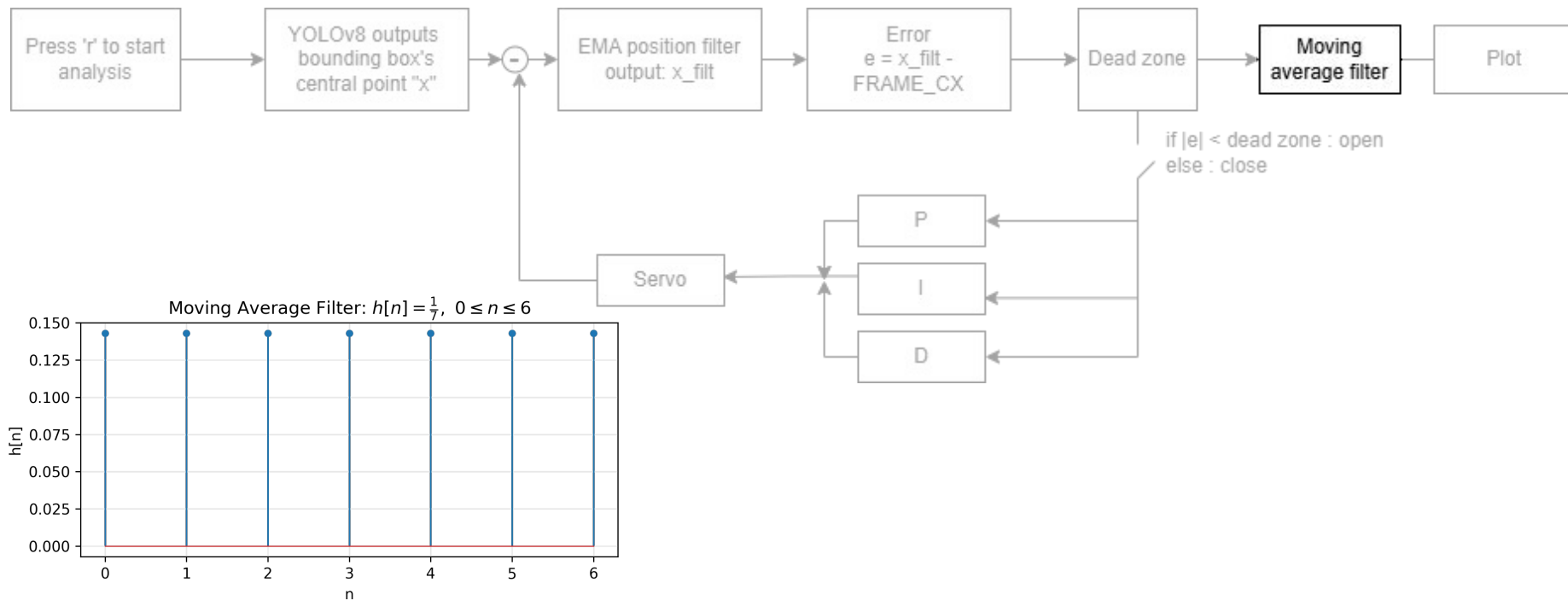
$$h[n] = \alpha(1 - \alpha)^n, \quad n \geq 0$$

$$h[n] = 0.2(0.8)^n, \quad n \geq 0$$



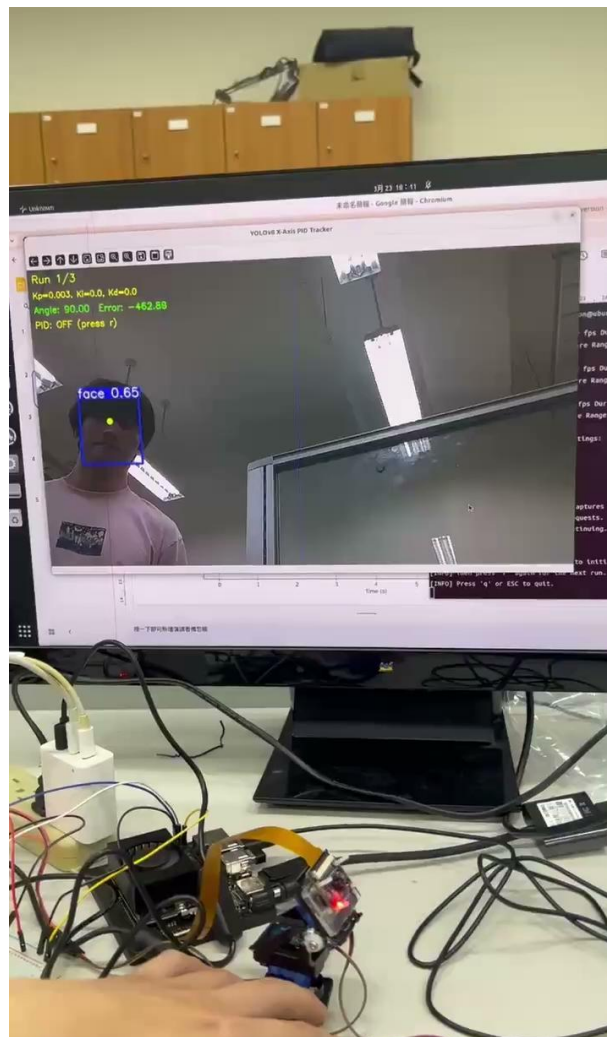


Moving Average Filter



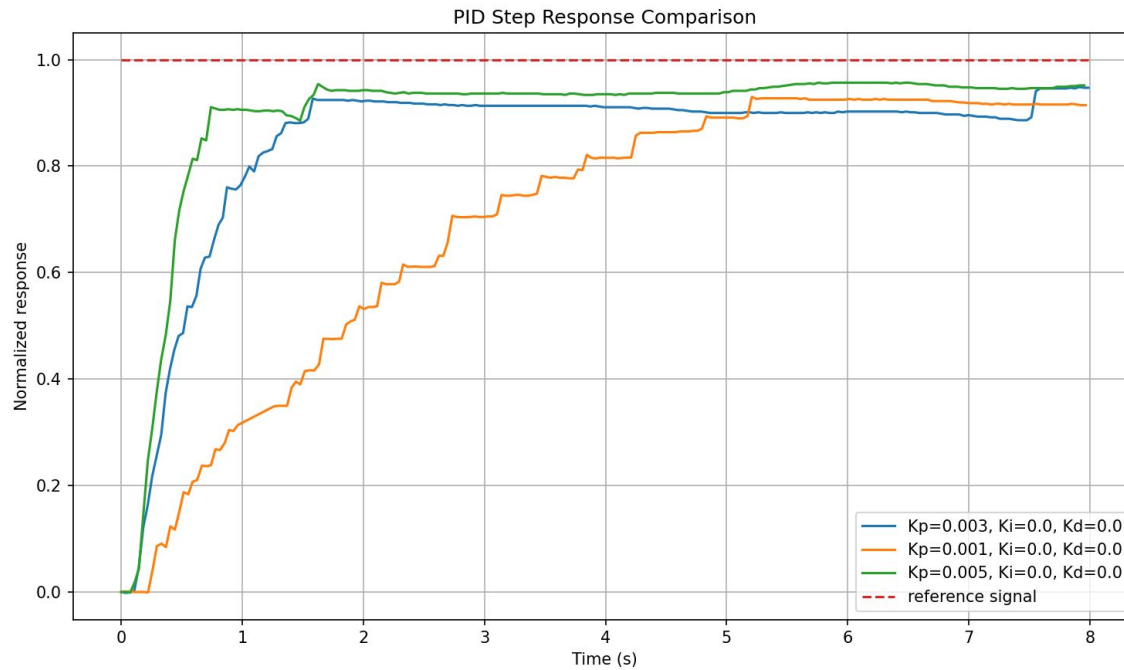


Demonstration





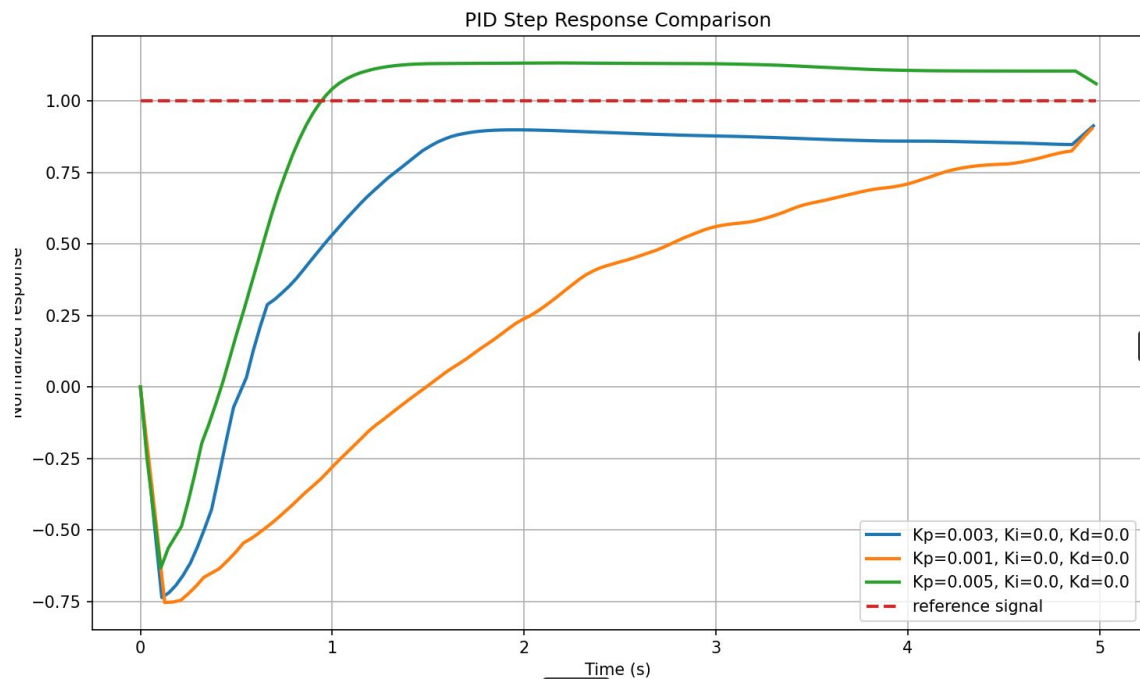
Unfiltered Data





Filtered Data

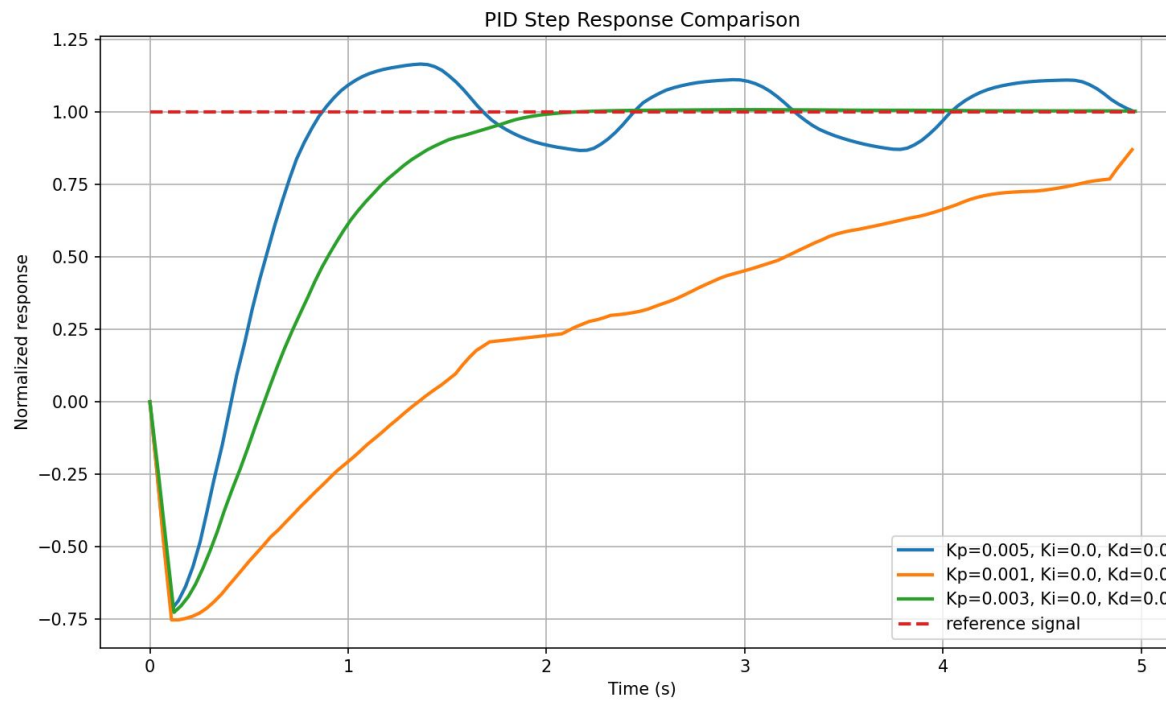
dead zone:40 duration:5s





Filtered Data

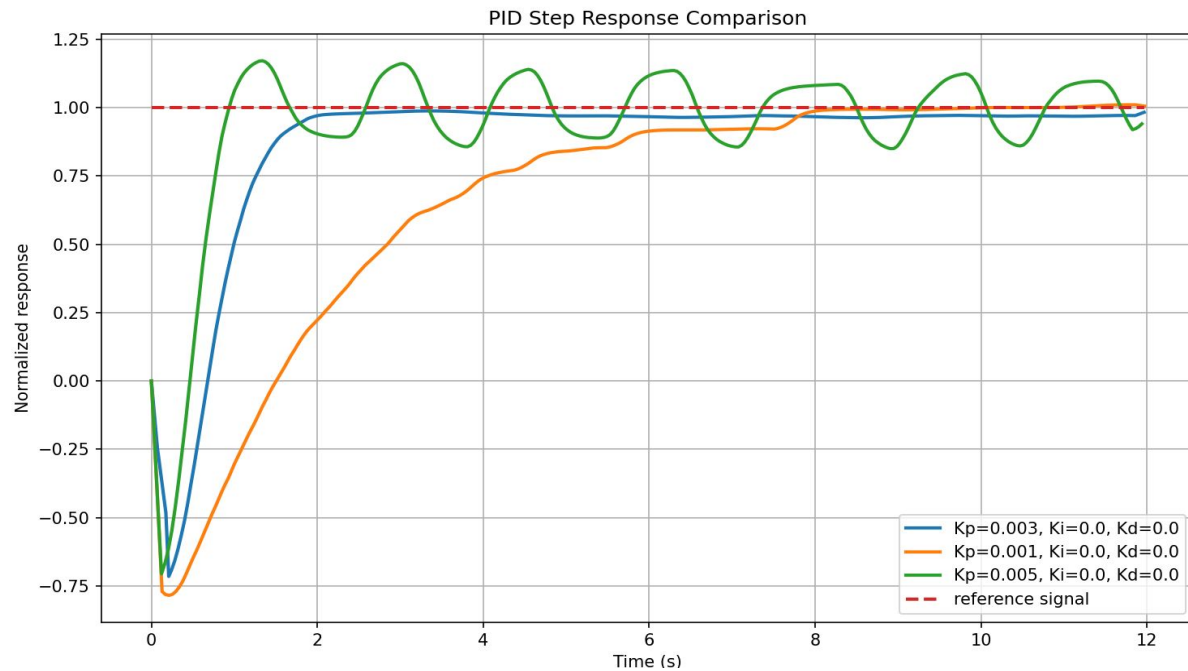
dead zone:10 duration:5s





Filtered Data

dead zone:10 duration:12s



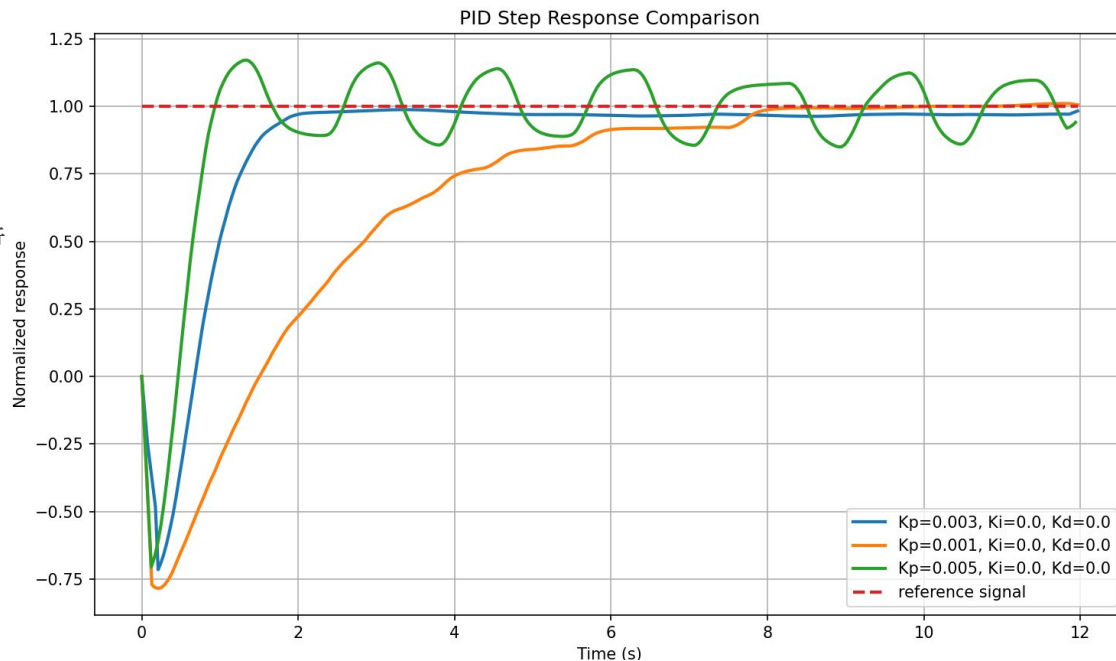


Conclusion

在 $K_p=0.005$ 時，系統雖然能最快到達 setpoint，但會產生持續震盪，無法穩定收斂。

相較之下， $K_p=0.003$ 雖然不是最快，但其暫態響應明顯優於 $K_p=0.001$ ，且能維持較好的穩定性。

綜合考量下， $K_p=0.003$ 在速度與穩定性之間取得較佳平衡。





Future Plan

本次Meeting進度：

- ✓ 成功處理套件不相容的問題。
- ✓ Code可以順利執行，成功利用 Jetson Orin Nano的GPIO控制伺服馬達。
- ✓ 成功在real-time情境下實現人臉PID追蹤(x-axis)。
- ✓ 分析不同PID參數對於追蹤人臉的影響。

未來計畫：

- 完成兩軸(x、y)之PID控制。
- 訓練 rolling shutter pattern 或其他可辨認船隻之模型。
- 在Free space下能real time追蹤到Pattern。